Conditional generation of non-Euclidean data: diffusion models and flow matching

Literature review and contribution assessment

Prepared for Yidong Zhou | I-2026-018

Evidence cutoff: October 31, 2026

Abstract

Conditional generation of non-Euclidean responses is an established and rapidly expanding area. Direct precedents include SPD matrices given mixed continuous predictors, correlation matrices given market variables, rotations and poses given images, molecular conformations given graphs or sequences, crystals given physical properties, and distribution-valued responses given biological context. The literature also contains conditional graph and tree generators, function-space methods, and sampling procedures conditioned on partial observations. Consequently, a distinction based only on adding continuous covariates to an existing geometric flow is not supported. The substantive questions concern the target conditional law, preservation of the relevant object geometry, estimation from finite paired data, and calibration of generated variation. This review separates those questions, examines 42 primary works spanning methods and their nearest theoretical comparators, and identifies the implications for a conditional FréchetFlow extension. The strongest remaining direction is a specific statistical or methodological result beyond routine conditioning, evaluated in the original response metric against both native geometric models and generators sharing the same representation and decoder.

1 Scope and the meaning of conditional generation

The primary target is a probability kernel $P_x = \mathcal{L}(Y \mid X = x)$, where $Y \in (\mathcal{M}, d)$ is an entire random object and X is an observed covariate. A fitted model should generate independent realizations of whole objects at a supplied x. Covariates can be continuous vectors, categories, images, sequences, other objects, or partially observed responses. The central distinction is between conditioning on information available at generation time and conditioning on an endpoint solely to construct a training loss.

Three mechanisms recur. *Direct conditional learning* fits a score, denoiser, vector field, or transition rate using paired (X, Y) observations. *Guidance* modifies a generator using a property predictor or conditional/unconditional model combination. *Conditional sampling from a learned prior* combines an unconditional generator with an observation likelihood or a completion constraint. These mechanisms can serve similar applications, but their training requirements and guarantees differ. In particular, stronger guidance can deliberately favor property satisfaction over faithful sampling from the original conditional law.

The response space also matters. A smooth manifold provides tangent spaces and geometric differential operators. Graphs, trees, and variable-cardinality sets require discrete or hybrid constructions. Distributions are objects in a space such as $\mathcal{P}_2(\mathcal{M})$. Hilbert-valued functions are an important adjacent case, although Hilbert geometry is linear rather than intrinsically curved. A Euclidean coordinate or latent representation of a shape does not, by itself, preserve a specified shape-space metric.

For distribution-valued responses, there are two probability levels. If Y is a cell-population distribution, P_x is a *law over population distributions*. A model that generates individual cells from one distribution at each x need not represent variation among populations under the same conditions. Likewise, a deterministic map from one population to another predicts one distribution-valued object, rather than a nondegenerate conditional law over possible objects.

The search covered general scholarly web discovery, arXiv, OpenReview, PMLR, NeurIPS and ICLR proceedings, CVF, journal and institutional repositories, Crossref metadata, and author code repositories. It combined object-specific terminology, backward and forward chaining, and author/venue searches with the October 15 review. The retained set has 42 distinct primary works; read depth and exact source locations are recorded individually in the accompanying evidence table. This is a focused research review, not an exhaustive systematic census. Recent preprints are identified as such, and proofs and experiments have not been independently reproduced.

2 Why conditioning is straightforward in both model families

Suppose an existing method maps an object to coordinates $Z = \phi(Y) \in \mathbb{R}^m$ and decodes terminal coordinates with D. For a linear flow-matching path, draw a paired observation (X, Y), independently draw $U \sim N(0, I_m)$ and $T \sim \mathrm{Unif}(0, 1)$, and set

$$Z_T = (1 - T)U + T\phi(Y), \qquad \mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}\big[\|v_\theta(T, Z_T; X) - \{\phi(Y) - U\}\|^2\big].$$

At generation, fix x, draw fresh U, solve $\dot{z}_t = v_\theta(t, z_t; x)$ with $z_0 = U$, and return $D(z_1)$. The network input contains the interpolated state Z_T, not the observed endpoint $\phi(Y)$. Keeping the observed X and Y paired is essential.

A conditional diffusion model has the same architectural principle. For a forward noising schedule, form $\widetilde{Z}_t = \alpha_t\phi(Y) + \sigma_t U$ and, for example, minimize

$$\mathcal{L}_{\mathrm{diff}}(\theta) = \mathbb{E}\big[w(T)\|\epsilon_\theta(T, \widetilde{Z}_T; X) - U\|^2\big].$$

Here forward time moves toward noise; generation reverses that process while holding x fixed. In a native manifold construction, the score or velocity instead takes values in the tangent space at the current response state. The covariate remains an input to that field. Direct conditional denoisers already appear in geometric molecular generation, while conditional transport theory formalizes the corresponding fixed-covariate flow construction (Hoogeboom et al., 2022; Kerrigan et al., 2024a).

Thus, replacing a categorical embedding with a continuous-covariate encoder usually requires modest changes once the underlying response generator exists. A particular implementation may also need a different base distribution, splitting rule, or conditioning architecture: DiffeoCFM's released implementation uses class-specific Gaussian bases, one-hot labels, and stratified splits. These implementation choices matter for a fair comparison, but do not establish a new conditional-generation principle. Learning accurate laws from finite pairs, choosing a valid geometry, and demonstrating useful conditional variation remain substantive tasks.

3 The closest matrix-valued precedents

SPD-DDPM already addresses vector-covariate conditional generation. The SPD-DDPM method of Li et al. (2024), published at AAAI 2024, learns $p(Y \mid X)$ for SPD responses with a conditional SPD denoising network. Its taxi application uses 13 predictors, including continuous travel summaries

and weather measurements such as temperature, humidity, wind speed, pressure, and precipitation. For each test predictor, it generates 20 matrices and evaluates their Riemannian center. This is a direct precedent for continuous-covariate object generation, although its conditional evaluation primarily addresses mean prediction rather than the full generated law. The public training code passes the predictor vector to the denoiser. A specific algebraic concern in the inspected preprint is recorded in the evidence notes; its conditional modeling capability does not depend on accepting every displayed derivation.

Financial correlation diffusion also predates the proposed extension. Kubiak et al. (2024) condition a DDPM on continuous interest-rate and equity volatility. The response is an empirical financial correlation matrix, and the two conditions enter the denoising network through embeddings. This establishes continuous conditioning in the application, but the model operates on matrix entries and subsequently enforces symmetry and unit diagonal. Those operations alone do not guarantee positive semidefiniteness. Its entrywise distributional summaries also differ from comparing laws of whole matrices in a prescribed matrix metric.

DiffeoCFM supplies a strong geometric FM baseline. Collas et al. (2025) use global diffeomorphisms and pullback geometry for covariance and correlation matrices. Training becomes Euclidean conditional FM in transformed coordinates, with valid matrices recovered through the inverse map. The demonstrations principally use diagnostic or motor-imagery classes; continuous variables such as age are discussed as a natural extension. The appropriate conclusion is that continuous-covariate evaluation is limited *in that paper*. It is not a literature-wide gap, nor a barrier to adapting its network. Its induced pullback geometry should also be distinguished from a metric independently selected for the scientific problem.

CorrFlow directly combines continuous conditioning with valid correlations. Marti (2026) uses generalized Fisher coordinates and a permutation-equivariant graph velocity. The principal financial study conditions on a market-regime label and a continuous standardized portfolio-performance score. Its mathematical contribution concerns valid correlation matrices and equivariance. The main empirical comparison changes geometry, architecture, and objective together, while its categorical DiffeoCFM comparison uses a separate protocol with unequal training budgets. These results support the complete construction under the studied conditions; they do not isolate a universal advantage of FM over diffusion.

For matrix responses, the nearest comparison set therefore contains both conditional diffusion and conditional FM, including continuous predictors. A follow-up needs to establish something beyond the presence of covariates: improved conditional distributional accuracy, a different scientifically necessary geometry, a fitted estimator theorem, or a meaningful advantage in a well-defined data regime.

4 Distribution-valued responses and population generation

Wasserstein Flow Matching models laws over distributions. Haviv et al. (2025) develop flows for Gaussian distributions under Bures–Wasserstein geometry and more general distributions represented by point clouds. Its organ, cellular subtype, and shape-conditioning experiments directly generate whole objects. Categorical conditioning dominates the demonstrations. The finite-dimensional Gaussian construction and the more formal general Wasserstein derivation have different mathematical status; neither should be summarized as a finite-sample conditional-law estimation theorem for arbitrary covariates.

RWEFM broadens both the domain geometry and the conditioning examples. Haviv et al. (2026) study measures on Riemannian manifolds. It generates cell-population distributions in a spherical latent space under healthy/diseased labels and generates protein-specific torsion-angle distributions

on a torus conditional on sequence embeddings. In the latter application, the object pools angle pairs across residues and simulation frames; it is not a full joint distribution of complete protein structures. Its population flow and entropic-transport results provide mathematical structure, while finite-sample estimation of the learned conditional outer law remains a separate question.

PerturbDiff is especially close to an embedding-based proposal. Yuan et al. (2026) explicitly make a perturbed population distribution random and model its law given a control population, cellular context, and perturbation. It embeds distributions into an RKHS, formulates conditional diffusion there, and implements finite-batch denoising with a transformer and an MMD-plus-MSE loss. Practical noising uses a first-order feature-map approximation, and sampling takes place in cell space. Drug dose, when included, is discretized into a learned vocabulary rather than treated as a freely varying continuous covariate. Its biological prediction evaluations do not establish calibration of the outer law. Nevertheless, the central idea of conditional diffusion on representations of distribution-valued objects is explicit prior art.

Two nearby lines sharpen the target distinction. Atanackovic et al. (2025) condition transport on an embedding of the initial population and treatment, learning a population evolution rule that generalizes across patients. A single predicted target distribution is not automatically a conditional distribution *over* targets. Fishman et al. (2026) learn transport conditioned on source and target population embeddings, then fit a target-embedding predictor for deployment. Its stochastic predictor uses an energy-score loss and induces a mixture over transported populations. This is a close two-stage representation-and-generation precedent even though its outer conditional generator is not itself FM. Its embedding-loss central limit theorem fixes model parameters and addresses finite within-population sampling, rather than end-to-end estimation of the fitted conditional law.

Piening et al. (2026) provide a relevant theoretical refinement: nested outer and inner transport plans generate flows over measures and yield a least-squares velocity characterization. The October 26 version is titled *Generalized Wasserstein Flow Matching: Nested Couplings for Point Cloud Generation*. Its endpoint-conditioned construction is not counted here as a demonstrated external-covariate model. It does, however, narrow the scope for claiming that general measure-valued FM is mathematically unexplored.

For a statistical project, the response level must be stated before selecting a baseline: an individual cell, a population distribution, or a random population distribution conditional on observed information. With empirically observed populations, one must also distinguish biological variation among populations from sampling variation within each finite cell batch. A theorem addressing that observation model would answer a more specific question than introducing covariates to a velocity network.

5 Other non-Euclidean response families

5.1 Rotations, poses, and action trajectories

Image-conditioned pose generation is a direct example of noncategorical covariates with a manifold response. Hsiao et al. (2024) model a distribution on $SE(3)$ from one RGB image to represent pose ambiguity induced by symmetry and occlusion. The conditional task is central to the construction, rather than an incidental class label.

For FM, Ding et al. (2025) learn observation-conditioned action trajectories on product Riemannian spaces. Visual and state information enter the vector field, and the generated trajectory is used in a receding-horizon policy. Its stable extension supplies attraction to target support. Such a stability result differs from recovering the correct probabilities within that support. The policy experiments

are evidence for an effective conditional generator in robotics, not a general statistical comparison of conditional-law estimators.

Zhong et al. (2026) extend average-velocity modeling to manifolds through parallel transport and demonstrate one-step class-conditioned generation on an $SO(3)$ mixture. This is relevant to sampling efficiency, but its preliminary categorical experiment does not establish a new continuous-covariate regression result.

5.2 Molecular conformations, proteins, and crystals

Jing et al. (2022) perform diffusion on a hypertorus of molecular torsion angles conditional on molecular structure and local geometry. Corso et al. (2023) generate ligand poses conditional on a protein and ligand using translation, rotation, and torsional degrees of freedom. These are established examples of structured predictors and genuinely geometric response constraints. Docking quality after confidence ranking should be distinguished from calibration of all generated poses.

On the FM side, FOLDFlow-2 (called FoldFlow++ in the NeurIPS 2024 record) conditions backbone generation on amino-acid sequences, with rotation and translation losses for residue frames and additional structure-inpainting applications (Huguet et al., 2024). RFdiffusion supplies a major diffusion precedent for motif-conditioned scaffolding and target-dependent protein design (Watson et al., 2023). Neither biological usefulness nor designability alone establishes the full physical conditional equilibrium law.

Continuous property conditioning is also established in geometric and hybrid object generation. Equivariant diffusion generates 3D molecules under supplied molecular properties (Hoogeboom et al., 2022); its representation combines atomic coordinates and categorical atom types while respecting Euclidean symmetries. MatterGen jointly generates atomic species, periodic coordinates, and a crystal lattice, with fine-tuning for desired chemistry, symmetry, and continuous physical properties (Zeni et al., 2025). These applications demonstrate substantial conditional-generation capability. Their inverse-design objectives emphasize finding suitable objects, which can differ from estimating an observational conditional distribution without selection or guidance.

5.3 Graphs and phylogenetic trees

DiGress uses discrete graph diffusion and a noisy-graph property regressor to guide samples toward specified graph-level properties (Vignac et al., 2023). Its guidance approximation is distinct from training a fully covariate-conditioned reverse model. DeFoG uses discrete FM and includes direct conditional training with classifier-free guidance; its digital-pathology experiment conditions on low versus high tertiary lymphoid structure content (Qin et al., 2025). The relevant geometry is combinatorial, with transition rates replacing ordinary tangent-vector ODEs. BWFlow adds a Bures–Wasserstein construction based on graph Markov random fields, but is retained here as a geometric graph-generation comparator rather than a verified external-covariate experiment (Jiang et al., 2026).

For phylogenetics, PhylaFlow combines continuous branch-length dynamics with topology changes in BHV tree space and includes sequence-conditioned generation (Ektefaie et al., 2026). Its joint conditional experiment should be separated from other experiments that fit one model per dataset. The intended use includes posterior proposals and refinement, so success of those procedures does not by itself establish an accurately calibrated full tree law.

Baron et al. (2026) provide a recent discrete-diffusion counterpart. It trains on simulated sequence–tree pairs and learns top-down, middle-out, or bottom-up reverse kernels for tree topology. It compares posterior frequencies and empirical calibration with MCMC on real and synthetic datasets. The response

excludes branch lengths, and out-of-distribution behavior depends on the evolutionary training model. Thus, neither conditional tree generation nor empirical posterior-calibration assessment is an untouched direction; a proposed result should specify which topology/length target and which statistical guarantee it adds.

5.4 Fields, functions, point processes, and latent shapes

Geometric Neural Diffusion Processes constructs distributions over fields with symmetry and develops conditional Langevin sampling from context observations (Mathieu et al., 2023). The distinction between a curved input domain and a curved value space is important: Euclidean-valued functions defined on a sphere are not automatically sphere-valued responses. The paper also treats spherical codomains, making it directly relevant to geometric field generation.

Functional Flow Matching develops measure paths in Hilbert spaces and demonstrates conditioning on observed function values through conditional training and a modified sampler (Kerrigan et al., 2024b). COT-FM supplies a more explicit conditional transport theory on separable Hilbert spaces and an infinite-dimensional Bayesian inverse-problem application (Kerrigan et al., 2024a). On the diffusion side, ∞-Brush incorporates external conditioning through a function-space cross-attention construction (Le et al., 2024). These works establish important infinite-dimensional precedents, while relying on linear-space structure unavailable in a general metric space.

Point Set Diffusion generates variable-cardinality point configurations and conditions on observed points under a spatial or temporal mask (Lüdke et al., 2025). EDITPP similarly uses an unconditionally trained edit-rate process for conditional temporal event generation (Lüdke et al., 2026). Both support forecasting or completion. They should not be described as external subject-covariate regression experiments merely because their samplers are conditional.

Michelangelo illustrates a common representation route: conditional diffusion in a learned latent space followed by a shape decoder, with text or images as conditions (Zhao et al., 2023). Such pipelines are direct conditional object generators, but an intrinsic error bound in a chosen shape metric requires an additional representation/decoder argument. DUALFLOW is an older manifold-to-manifold conditional generator, retained as broader context; it is a normalizing-flow method, not diffusion or flow matching (Zhen et al., 2021).

6 Comparison by actual conditioning information

Table 1 collects the most consequential precedents. "Direct" means external information enters a model trained on paired conditions and responses. It does not imply a fitted statistical consistency theorem.

Table 1: What is generated, and what actually conditions it.

Work	Response	External information	Mechanism / scope
SPD-DDPM (2024)	SPD matrices	13 mixed travel/weather predictors	Direct diffusion; continuous predictors
Financial DDPM (2024)	Correlation matrices	Two continuous volatility measures	Direct ambient diffusion; matrix repair
DiffeoCFM (2025)	Covariance/correlation matrices	Diagnosis and task classes	Direct FM; class-based implementation

Table 1 continued

Work	Response	External information	Mechanism / scope
CorrFlow (2026 preprint)	Correlation matrices	Regime plus continuous performance score	Direct FM in exact coordinates
WFM (2025)	Whole probability measures	Organ/subtype/shape labels	Direct measure-valued FM
RWEFM (2026 preprint)	Measures on manifolds	Tissue status; protein sequence embedding	Direct FM; spherical/torus applications
PerturbDiff (2026 preprint)	Cell-population distributions	Control population, context, perturbation	Direct diffusion formulation; dose discretized
Hsiao et al. (2024)	$SE(3)$ pose	RGB image	Direct score diffusion
RFMP (2025)	Action trajectories	Visual and state observations	Direct Riemannian FM policy
Torsional diffusion (2022)	Molecular torsion angles	Molecular graph/local geometry	Direct hypertorus diffusion
DiffDock (2023)	Ligand pose	Protein and ligand structures	Direct product-manifold diffusion
FOLDFlow-2 / ++ (2024)	Protein backbone	Amino-acid sequence	Direct geometric FM; also inpainting
RFdiffusion (2023)	Protein structures	Motif/target/design constraints	Conditional diffusion/design
EDM (2022)	3D molecules	Continuous molecular properties	Direct equivariant diffusion
MatterGen (2025)	Periodic crystals	Chemistry, symmetry, scalar properties	Conditional adaptation/guidance
DiGress (2023)	Graphs	Desired graph properties	Regressor-guided diffusion
DeFoG (2025)	Graphs	Low/high TLS label	Direct discrete FM plus guidance
PhylaFlow (2026 preprint)	Topology and branch lengths	Sequence information	Hybrid FM; proposal/refinement uses
Baron et al. (2026 preprint)	Tree topology	Sequence alignment	Direct discrete diffusion; simulated pairs
GNDP (2023)	Geometric fields	Observed context values/locations	Conditional Langevin sampling
FFM / COT-FM (2024)	Functions	Observed values / inverse measurements	Conditional functional flows
Point Set Diffusion / EDITPP	Point configurations	Observed region or subsequence	Inference-time mask conditioning

The continuous-covariate distinction therefore fails as a broad novelty claim. Even within matrices, SPD-DDPM and financial DDPM predate DiffeoCFM, and CorrFlow combines continuous conditioning with exact correlation coordinates. Structured inputs are equally established in poses, molecular geometry, protein generation, and distributions on manifolds. Categorical experiments in one paper do not imply that the method class is restricted to categories.

7 What is established theoretically

7.1 Geometry and population dynamics

Riemannian Score-Based Generative Modelling and Riemannian Diffusion Models develop diffusion on smooth manifold states (De Bortoli et al., 2022; Huang et al., 2022). Riemannian Flow Matching develops tractable geometric probability paths and field regression (Chen and Lipman, 2024). These foundations require more than a distance oracle: their constructions use differential structure, geometry-aware paths, or associated operators. Diffeomorphic coordinates can remove some computational difficulties, but only for spaces admitting the required representation.

Generator Matching unifies flow, diffusion, and jump constructions through Markov generators (Holderrieth et al., 2025). Latent Process Generator Matching formalizes the projection of latent dynamics through measurable maps under domain, integrability, and uniqueness conditions (Billera et al., 2026). These are relevant to broad metric-space claims. Their generality does not automatically produce a practical sampler or a finite-sample conditional-law theorem for an arbitrary learned encoder and decoder.

For distributions, GWFM and RWEFM give flow/continuity-equation constructions on spaces of measures (Piening et al., 2026; Haviv et al., 2026). For graphs, DeFoG relates denoising/transition estimation error to sampling behavior (Qin et al., 2025). These are stronger than empirical demonstrations alone. They should still be separated from a theorem bounding the conditional law produced by a fitted model as the number of observed (X, Y) pairs increases.

7.2 Fitted conditional-law estimation

The conditional diffusion analysis of Tang et al. (2025) is the strongest directly relevant statistical diffusion precedent in the retained set. It studies conditional distributions of ambient Euclidean responses with smooth, potentially covariate-dependent manifold support. Theorem 2 bounds integrated conditional W_1 error with dependence on intrinsic predictor and response dimensions. The assumptions include controlled local packing of the predictor support, smooth local response parameterizations, and smooth positive conditional densities. Thus, the result reaches beyond a fixed response manifold or a categorical covariate, but does not cover arbitrary metric or singular object spaces. The published main text defers the fully formal response assumption to a supplement; the exact rate should not be reused without checking that complete statement.

The companion minimax study by Tang and Yang (2025) derives lower and upper bounds for distribution regression on fixed and covariate-dependent smooth manifolds, using a different estimator. It is a theoretical comparator even though it is not a diffusion/FM method. Together, these papers rule out a general claim that continuous-covariate non-Euclidean distribution estimation lacks statistical theory.

For a metric response, a natural target is

$$\mathcal{R}_p(\widehat{P}, P) = \left\{ \int W_{p,d}^p(\widehat{P}_x, P_x)\, P_X(dx) \right\}^{1/p},$$

under suitable moment conditions. A theorem should specify the actual trained estimator, representation, decoder, and sampling approximation. Assuming convergence of a coordinate-space conditional generator and then applying an existing deterministic decoding bound at each x yields a useful transfer corollary, but leaves the fitted conditional learning problem unresolved.

Conditional transport theory also identifies the correct coupling constraint. Chemseddine et al. (2025) show why ordinary Wasserstein agreement of joint laws need not control expected conditional Wasserstein

agreement. COT-FM develops covariate-preserving transport paths and triangular vector fields in Hilbert spaces (Kerrigan et al., 2024a). A proposed conditional OT or transport-stability result must be compared against these formulations rather than against unconditional FM alone.

7.3 Sampling accuracy and calibration

The Twisted Diffusion Sampler uses sequential Monte Carlo to condition a learned diffusion model, including a Riemannian protein-model extension (Wu et al., 2023). Its particle-limit guarantee targets the *fitted model's* conditional law. It does not establish that the fitted prior equals the scientific data law. This is a meaningful conditional-sampling theorem with a different asymptotic parameter from the number of training observations.

Xu et al. (2026) give polynomial-step sampling bounds on connected compact manifolds under bounded geometry and score-error conditions. The total-variation comparison is to an early-stopped, heat-smoothed target. It supplies current geometric sampling theory, rather than a paired-data conditional estimation rate or total-variation consistency for an arbitrary singular unsmoothed law.

Table 2 summarizes the distinctions needed when evaluating a new contribution.

Table 2: Different guarantees answer different questions.

Guarantee	Representative evidence	Remaining statistical question
Valid objects / symmetry	DiffeoCFM, CorrFlow, equivariant molecular models	Are conditional probabilities and variability accurate?
Population flow identity	RFM, WFM/GWFM/RWEFM, Generator Matching, COT-FM	How accurately is the relevant field or generator learned from finite pairs?
Sampling error given learned quantities	DeFoG; Riemannian diffusion bounds	How do estimation, early stopping, and discretization errors combine?
Exact conditioning in a particle limit	Twisted Diffusion Sampler	How accurate is the fitted unconditional model and the observation model?
Fitted conditional-law rate	Tang–Lin–Yang; related manifold minimax theory	What changes for a different object metric, singular geometry, or noisy object observation?
Empirical conditional success	Pose, policy, protein, material, graph and tree studies	Does the chosen metric assess the entire intended conditional law?

8 Implications for a conditional FréchetFlow project

Feasibility is high; routine conditioning is already established. Adding X to the current velocity network is a natural extension, and it could make the framework more useful. The evidence does not support presenting that step, the use of continuous predictors, or generic conditional generation of non-Euclidean objects as the central novelty of a separate paper. The same conclusion applies to switching the coordinate-space generator from FM to diffusion without another substantive change.

A distinct contribution needs a precise unresolved target. One possible target is a fitted conditional-law bound in the original response metric for a specified nonsmooth object space, with explicit response complexity, covariate regularity, representation budget, and decoder error. The difficult part must exceed applying the existing unconditional transfer bound pointwise. A second

is a scientifically motivated observation model for distribution-valued responses observed through finite samples, separating object-level variation from within-object noise. A third is a conditioning-aware representation or sampling method that demonstrably improves conditional fidelity or cost in a named setting. These are candidate questions to investigate, not verified literature gaps; the tree, measure-valued, and representation precedents above are immediate comparisons.

The decisive empirical comparison should isolate the contribution. For matrices, include continuously conditioned native geometric FM and an appropriate conditional diffusion model. For distribution-valued responses, compare the relevant WFM/RWEFM or embedding-based conditional construction. Also compare conditional resampling and generators using the same encoder and decoder. This separates benefits of the representation, decoder, conditioning architecture, objective, and solver. Use common paired data, held-out conditions, training budgets, and sampling budgets; report both valid-object generation and distributional fidelity in the chosen object metric.

For continuous covariates, evaluating only pooled generated objects or predicted Fréchet means is insufficient to assess P_x. Useful diagnostics include held-out conditional distributional scores, calibration of meaningful object events, and simulations where the conditional law is known. Graph/topology probabilities and within-topology geometry may need separate diagnostics. With distribution-valued responses, evaluation should assess variation among generated distributions as well as accuracy within each one. These requirements follow from the statistical target and do not favor FM or diffusion in advance.

The literature supports choosing between FM and diffusion by the geometry, available construction, data regime, and measured accuracy–cost tradeoff. FM often makes a chosen path easy to train against; diffusion has mature noising and conditional-sampling constructions. Both accommodate external covariates. Task-specific comparisons such as CorrFlow or RFMP do not establish an across-domain ranking, especially when they change the backbone, coordinates, guidance, or training budget along with the objective.

9 Reading priorities and evidence limits

For the proposed statistical follow-up, the first reading group is SPD-DDPM, DiffeoCFM, CorrFlow, WFM, RWEFM, and PerturbDiff. They establish the nearest actual conditional capabilities. The second is Tang–Lin–Yang, COT-FM, conditional Wasserstein distances, and the manifold minimax paper, which define the theoretical comparison. The third depends on the application: PhylaFlow and the new discrete tree diffusion for trees, RFMP for actions, and DiGress/DeFoG for graphs.

The evidence archive contains exact queries, authoritative metadata, full-text files, section-level notes, and selected code inspections. Code was read to verify conditioning interfaces; no reviewed model was trained or run. The AAAI download endpoint returned an anti-bot page, so SPD-DDPM's arXiv full text and Crossref publication metadata were used. Some supplements and repository links remained unavailable; access limits are recorded individually. A preprint or a method's own reported experiment is evidence of a construction or reported result, not independent replication. Before external positioning, refresh any narrowly stated gap against the intended response space and theorem assumptions.

References

Lazar Atanackovic, Xi Zhang, Brandon Amos, Mathieu Blanchette, Leo J. Lee, Yoshua Bengio, Alexander Tong, and Kirill Neklyudov. Meta Flow Matching: Integrating Vector Fields on the Wasserstein Manifold. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/ebdb990471f653dffb425eff03c7c980-Abstract-Conference.html.

Ethan Baron, Alan N. Amin, and Andrew Gordon Wilson. Deep Generative Models for Phylogenetic Inference with Complex Evolutionary Processes, 2026. URL https://arxiv.org/abs/2610.19053. Preprint; version and metadata checked October 31, 2026.

Lukas Billera, Hedwig Nora Nordlinder, and Ben Murrell. Latent Process Generator Matching, 2026. URL https://arxiv.org/abs/2605.20547. Preprint; version and metadata checked October 31, 2026.

Jannis Chemseddine, Paul Hagemann, Gabriele Steidl, and Christian Wald. Conditional Wasserstein Distances with Applications in Bayesian OT Flow Matching. *Journal of Machine Learning Research*, 26(141):1–47, 2025. URL https://jmlr.org/papers/v26/24-0586.html.

Ricky T. Q. Chen and Yaron Lipman. Flow Matching on General Geometries. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2302.03660.

Antoine Collas, Ce Ju, Nicolas Salvy, and Bertrand Thirion. Riemannian Flow Matching for Brain Connectivity Matrices via Pullback Geometry. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-1993. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/5616112a0120c15bf7d47a6bccc21bc3-Abstract-Conference.html.

Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. DiffDock: Diffusion Steps, Twists, and Turns for Molecular Docking. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.01776.

Valentin De Bortoli, Emile Mathieu, Michael Hutchinson, James Thornton, Yee Whye Teh, and Arnaud Doucet. Riemannian Score-Based Generative Modelling. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://arxiv.org/abs/2202.02763.

Haoran Ding, Noémie Jaquier, Jan Peters, and Leonel Rozo. Fast and Robust Visuomotor Riemannian Flow Matching Policy. *IEEE Transactions on Robotics*, 41:5327–5343, 2025. doi: 10.1109/TRO.2025.3601293. URL https://arxiv.org/abs/2412.10855.

Yasha Ektefaie, Leo Cui, Shrey Jain, Marinka Zitnik, and Pardis Sabeti. PhylaFlow: Hybrid Flow Matching in Billera-Holmes-Vogtmann Tree Space for Phylogenetic Inference, 2026. URL https://arxiv.org/abs/2605.21859. Preprint; version and metadata checked October 31, 2026.

Nic Fishman, Gokul Gowri, Paolo L. B. Fischer, Marinka Zitnik, Omar Abudayyeh, and Jonathan Gootenberg. Distribution-Conditioned Transport, 2026. URL https://arxiv.org/abs/2603.04736. Preprint; version and metadata checked October 31, 2026.

Doron Haviv, Aram-Alexandre Pooladian, Dana Pe'er, and Brandon Amos. Wasserstein Flow Matching: Generative modeling over families of distributions. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 22238–22258, 2025. URL https://proceedings.mlr.press/v267/haviv25a.html.

Doron Haviv, Edward De Brouwer, Rishabh Anand, Rex Ying, Aïcha Bentaieb, Gabriele Scalia, and Hector Corrada Bravo. When Riemann flows with Wasserstein: Generative Modeling of Probability Distributions on Manifolds, 2026. URL https://arxiv.org/abs/2609.25659. Preprint; version and metadata checked October 31, 2026.

Peter Holderrieth, Marton Havasi, Jason Yim, Neta Shaul, Itai Gat, Tommi Jaakkola, Brian Karrer, Ricky T. Q. Chen, and Yaron Lipman. Generator Matching: Generative modeling with arbitrary Markov processes. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2410.20587.

Emiel Hoogeboom, Victor Garcia Satorras, Clément Vignac, and Max Welling. Equivariant Diffusion for Molecule Generation in 3D. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 8867–8887, 2022. URL https://proceedings.mlr.press/v162/hoogeboom22a.html.

Tsu-Ching Hsiao, Hao-Wei Chen, Hsuan-Kung Yang, and Chun-Yi Lee. Confronting Ambiguity in 6D Object Pose Estimation via Score-Based Diffusion on SE(3). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 352–362, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Hsiao_Confronting_Ambiguity_in_6D_Object_Pose_Estimation_via_Score-Based_Diffusion_CVPR_2024_paper.html.

Chin-Wei Huang, Milad Aghajohari, Avishek Joey Bose, Prakash Panangaden, and Aaron Courville. Riemannian Diffusion Models. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://arxiv.org/abs/2208.07949.

Guillaume Huguet, James Vuckovic, Kilian Fatras, Eric Thibodeau-Laufer, Pablo Lemos, Riashat Islam, Cheng-Hao Liu, Jarrid Rector-Brooks, Tara Akhound-Sadegh, Michael Bronstein, Alexander Tong, and Avishek Joey Bose. Sequence-Augmented SE(3)-Flow Matching For Conditional Protein Generation. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-1039. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/39ca8893ea38905a9d2ffe786e85af0f-Abstract-Conference.html.

Keyue Jiang, Jiahao Cui, Xiaowen Dong, and Laura Toni. Bures-Wasserstein Flow Matching for Graph Generation. In *International Conference on Learning Representations*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/e3b1291b7529063172d927588d2b03a1-Abstract-Conference.html.

Bowen Jing, Gabriele Corso, Jeffrey Chang, Regina Barzilay, and Tommi Jaakkola. Torsional Diffusion for Molecular Conformer Generation. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://arxiv.org/abs/2206.01729.

Gavin Kerrigan, Giosue Migliorini, and Padhraic Smyth. Dynamic Conditional Optimal Transport through Simulation-Free Flows. In *Advances in Neural Information Processing Systems*, volume 37, 2024a. doi: 10.52202/079017-2968. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/aa1d8cf866c4a684ef2e066dd200f8e4-Abstract-Conference.html.

Gavin Kerrigan, Giosue Migliorini, and Padhraic Smyth. Functional Flow Matching. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 3934–3942, 2024b. URL https://proceedings.mlr.press/v238/kerrigan24a.html.

Szymon Kubiak, Tillman Weyde, Oleksandr Galkin, Daniel Philps, and Ram Gopal. Denoising Diffusion Probabilistic Model for Realistic Financial Correlation Matrices. In *Proceedings of the 5th ACM International Conference on AI in Finance*, pages 1–9, 2024. doi: 10.1145/3677052.3698640. URL https://openaccess.city.ac.uk/id/eprint/34037/.

Minh-Quan Le, Alexandros Graikos, Srikar Yellapragada, Rajarsi Gupta, Joel Saltz, and Dimitris Samaras. ∞-Brush: Controllable Large Image Synthesis with Diffusion Models in Infinite Dimensions. In *European Conference on Computer Vision*, 2024. URL https://arxiv.org/abs/2407.14709.

Yunchen Li, Zhou Yu, Gaoqi He, Yunhang Shen, Ke Li, Xing Sun, and Shaohui Lin. SPD-DDPM: Denoising Diffusion Probabilistic Models in the Symmetric Positive Definite Space. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(12):13709–13717, 2024. doi: 10.1609/aaai.v38i12.29276. URL https://arxiv.org/abs/2312.08200.

David Lüdke, Enric Rabasseda Raventós, Marcel Kollovieh, and Stephan Günnemann. Unlocking Point Processes through Point Set Diffusion. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2410.22493.

David Lüdke, Marten Lienen, Marcel Kollovieh, and Stephan Günnemann. Edit-Based Flow Matching for Temporal Point Processes. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2510.06050.

Gautier Marti. CorrFlow: Exact and Permutation-Equivariant Flow Matching on the Correlation Elliptope, 2026. URL https://arxiv.org/abs/2610.13606. Preprint; version and metadata checked October 31, 2026.

Emile Mathieu, Vincent Dutordoir, Michael J. Hutchinson, Valentin De Bortoli, Yee Whye Teh, and Richard E. Turner. Geometric Neural Diffusion Processes. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/a797c2d2e0c1fdabf4d1ab8cd0b465c6-Abstract.html.

Moritz Piening, Richard Duong, and Gabriele Steidl. Generalized Wasserstein Flow Matching: Nested Couplings for Point Cloud Generation, 2026. URL https://arxiv.org/abs/2605.08424v3. Preprint; version and metadata checked October 31, 2026.

Yiming Qin, Manuel Madeira, Dorina Thanou, and Pascal Frossard. DeFoG: Discrete Flow Matching for Graph Generation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 50269–50326, 2025. URL https://proceedings.mlr.press/v267/qin25d.html.

Rong Tang and Yun Yang. Minimax Optimal Rates for Regression on Manifolds and Distributions, 2025. URL https://arxiv.org/abs/2506.07504. Preprint; version and metadata checked October 31, 2026.

Rong Tang, Lizhen Lin, and Yun Yang. Conditional Diffusion Models are Minimax-Optimal and Manifold-Adaptive for Conditional Distribution Estimation. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a5e2853be2ff6218a8d87a5ce4b65062-Abstract-Conference.html.

Clement Vignac, Igor Krawczuk, Antoine Siraudin, Bohan Wang, Volkan Cevher, and Pascal Frossard. DiGress: Discrete Denoising diffusion for graph generation. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2209.14734.

Joseph L. Watson, David Juergens, Nathaniel R. Bennett, Brian L. Trippe, Jason Yim, Helen E. Eisenach, Woody Ahern, Andrew J. Borst, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Nikita Hanikel, Samuel J. Pellock, Alexis Courbet, William Sheffler, Jue Wang, Preetham Venkatesh, Isaac Sappington, Susana Vázquez Torres, Anna Lauko, Valentin De Bortoli, Emile Mathieu, Sergey Ovchinnikov, Regina Barzilay, Tommi S. Jaakkola, Frank DiMaio, Minkyung Baek, and David Baker. De novo design of protein structure and function with RFdiffusion. *Nature*, 620:1089–1100, 2023. doi: 10.1038/s41586-023-06415-8. URL https://www.nature.com/articles/s41586-023-06415-8.

Luhuan Wu, Brian L. Trippe, Christian A. Naesseth, David M. Blei, and John P. Cunningham. Practical and Asymptotically Exact Conditional Sampling in Diffusion Models. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2306.17775.

Xingyu Xu, Ziyi Zhang, Yorie Nakahira, Guannan Qu, and Yuejie Chi. Polynomial Convergence of Riemannian Diffusion Models, 2026. URL https://arxiv.org/abs/2601.02499. Preprint; version and metadata checked October 31, 2026.

Xinyu Yuan, Xixian Liu, Ya Shi Zhang, Zuobai Zhang, Hongyu Guo, and Jian Tang. PerturbDiff: Functional Diffusion for Single-Cell Perturbation Modeling, 2026. URL https://arxiv.org/abs/2602.19685. Preprint; version and metadata checked October 31, 2026.

Claudio Zeni, Robert Pinsler, Daniel Zügner, Andrew Fowler, Matthew Horton, Xiang Fu, Zilong Wang, Aliaksandra Shysheya, Jonathan Crabbé, Shoko Ueda, Roberto Sordillo, Lixin Sun, Jake Smith, Bichlien Nguyen, Hannes Schulz, Sarah Lewis, Chin-Wei Huang, Ziheng Lu, Yichi Zhou, Han Yang, Hongxia Hao, Jielan Li, Chunlei Yang, Wenjie Li, Ryota Tomioka, and Tian Xie. A generative model for inorganic materials design. *Nature*, 639:624–632, 2025. doi: 10.1038/s41586-025-08628-5. URL https://www.nature.com/articles/s41586-025-08628-5.

Zibo Zhao, Wen Liu, Xin Chen, Xianfang Zeng, Rui Wang, Pei Cheng, Bin Fu, Tao Chen, Gang Yu, and Shenghua Gao. Michelangelo: Conditional 3D Shape Generation based on Shape-Image-Text Aligned Latent Representation. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-3236. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/ea1a7f7bc0fc14142106a84c94c826d0-Abstract-Conference.html.

Xingjian Zhen, Rudrasis Chakraborty, Liu Yang, and Vikas Singh. Flow-based Generative Models for Learning Manifold to Manifold Mappings. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(12): 11042–11052, 2021. doi: 10.1609/aaai.v35i12.17318. URL https://ojs.aaai.org/index.php/AAAI/article/view/17318.

Zichen Zhong, Haoliang Sun, Yukun Zhao, Yongshun Gong, and Yilong Yin. Riemannian MeanFlow for One-Step Generation on Manifolds, 2026. URL https://arxiv.org/abs/2603.10718. Preprint; version and metadata checked October 31, 2026.